

Semester 1 · Academic Year 2026–2027

Exercise 3. Resistances in practice

(3 marks)

The typical input resistance of a digital voltmeter is:

- ☐ a few mΩ ☐ a few Ω ☐ a few kΩ ☐ 1 to 10 MΩ

To measure, with an ohmmeter, a resistor that is part of a circuit, you must:

- ☐ leave the circuit powered, so that a current flows through the resistor;
- ☐ switch the power supply off, but leave the resistor connected;
- ☐ disconnect the power supply and at least one terminal of the resistor;
- ☐ connect the ohmmeter in series with the resistor.

A resistor R carries a current i . The power dissipated by the Joule effect is:

- $$\square \quad Ri \qquad \square \quad R^2i \qquad \square \quad Ri^2 \qquad \square \quad u^2R$$

Exercise 4. Joule effect — a rear-window demister

(5 marks)

A car's rear-window demister is modelled by an ohmic resistor supplied by the car battery in the DC regime. Determine its resistance R , the current i it carries, then the energy E dissipated as heat in $\Delta t = 10 \text{ min}$.

$u = 12 \text{ V}$	$p = 72 \text{ W}$	$\Delta t = 10 \text{ min}$
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Marking

①	
0	1
☐	☐

②	
0	1
☐	☐

③	
0	1
☐	☐

④	
0	1
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⑤	
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— End of the test —